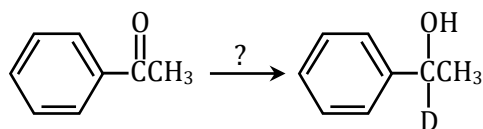




Alcohols and Ethers : RACE-01

Preparation of Alcohols

1. Which of the following reagents would carry out the following transformation ? (D = ^2H)



- (A) NaBD_4 in CH_3OH
(C) NaBD_4 in CH_3OD

- (B) LiAlH_4 , then D_2O
(D) LiAlD_4 , then D_2O

CAE188

2. $\text{C}_6\text{H}_5\text{CH}(\text{Cl})\text{COOCH}_2\text{CH}_3 + \text{CH}_3\text{COOH} \xrightarrow[\Delta]{\text{HCl}}$ (A) + (B)
(contains chlorine)
ethyl-2-chloro
2-phenyl acetate
(1 mole)
acetic acid

Product (A) and (B) respectively in the above reaction are :

- (A) $\text{C}_6\text{H}_5\text{CH}(\text{Cl})\text{COOH}$, $\text{CH}_3\text{COOCH}_2\text{CH}_3$ (B) $\text{C}_6\text{H}_5\text{CH}(\text{Cl})\text{COOEt}$, $\text{CH}_3\text{COOCH}_3$
(C) $\text{C}_6\text{H}_5\text{CH}(\text{Cl})\text{COOEt}$, CH_3COOH (D) $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$, CH_3COOH

CAE189

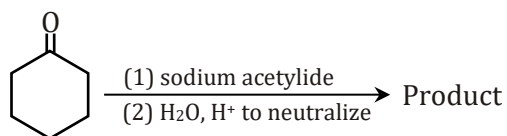
3. $\text{CH}_2=\text{CH}-\text{C}(=\text{O})\text{OCH}_3 + \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \xrightarrow[\text{bpt, } 145^\circ\text{C}]{\text{TsOH}, \Delta} \text{(A)} + \text{CH}_3\text{OH}$
methylacrylate n-butyl alcohol
bpt 81°C bpt 117°C

Product (A) in the above reaction are :

- (A) $\text{CH}_2=\text{CH}-\text{C}(=\text{O})\text{OCH}_2\text{CH}_2\text{CH}_3$ (B) $\text{CH}_2=\text{CH}-\text{C}(=\text{O})\text{OCH}(\text{Me})_2$
(C) $\text{CH}_2(\text{OCH}_3)-\text{CH}_2-\text{C}(=\text{O})\text{OCH}_3$ (D) $\text{CH}_3-(\text{CH}_2)_4-\text{C}(=\text{O})\text{OCH}_3$

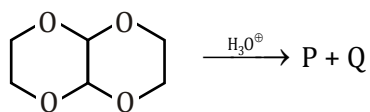
CAE190

4. Give the major organic product of the following reaction.



CAE191

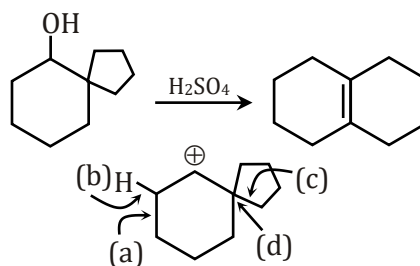
5. The acid catalysed hydrolysis products in the following reaction are -



CAE195

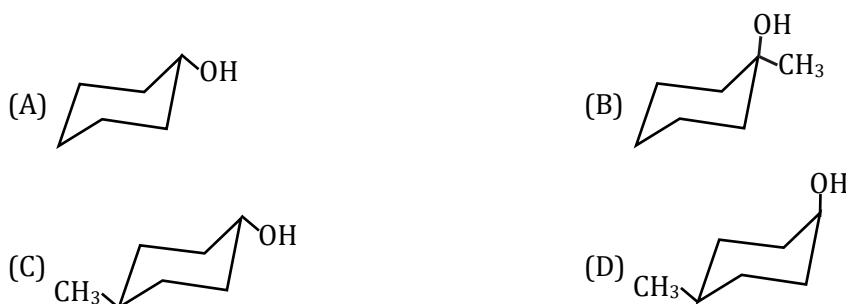
Reaction of Alcohols and Ethers

6. The following transformation involves a carbocation rearrangement. The carbocation is generated by protonation of the hydroxyl group, followed by the loss of water. which bond has to migrate in the carbocation to yield the product indicated (after the deprotonation) ?

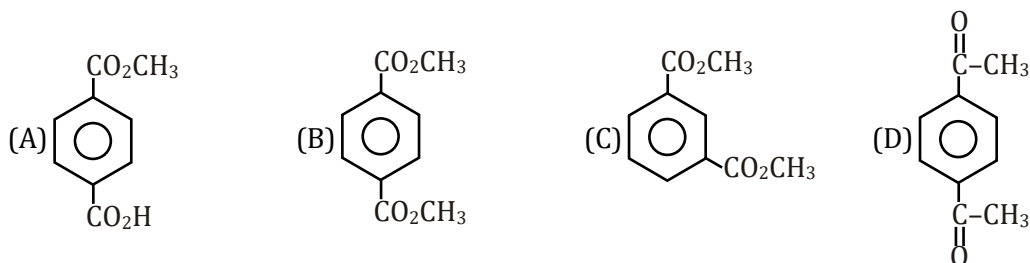
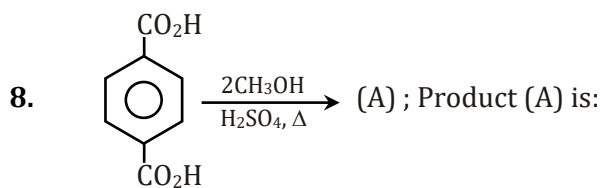


CAE176

7. Which of the following react with HBr at faster rate ?

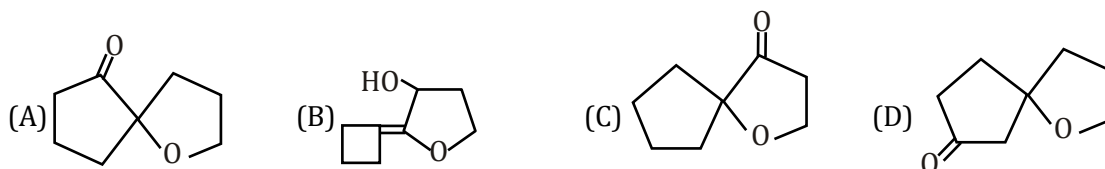
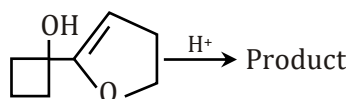


CAE177

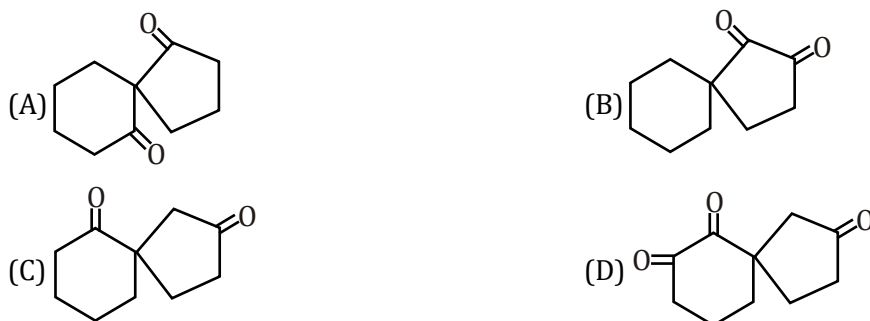
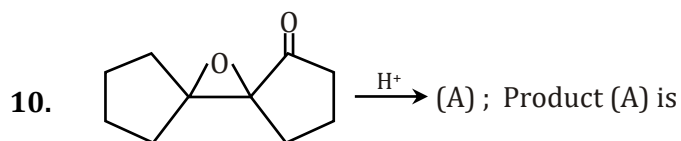


CAE178

9. Identify the major product,



CAE179



CAE180

ANSWER KEY

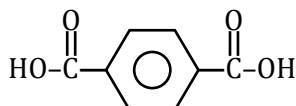
Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	A	C	C	C	B	B	A	A



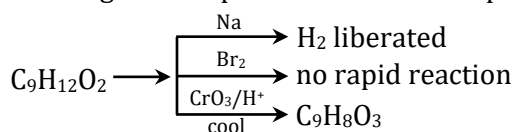
Alcohols and Ethers : RACE-02

Reaction of Alcohols and Ethers

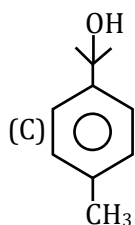
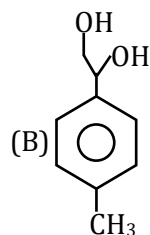
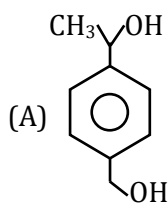
1. An optically active alcohol of formula $C_9H_{12}O_2$ produced the following compound when refluxed with $KMnO_4$.



The original compound showed these properties also :



What is Structure of (A) ?



(D) both (A) and (B)

CAE186

2. Which are not cleaved by HIO_4 ?

I : glycerol

II : glycol

III : 3-propenediol

IV : methoxy-2-propanol

(A) I,II,III,IV

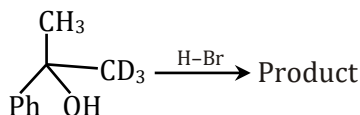
(B) I,II

(C) II,III

(D) III, IV

CAE187

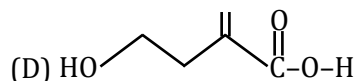
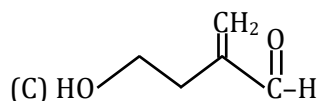
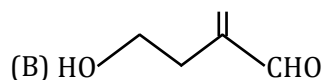
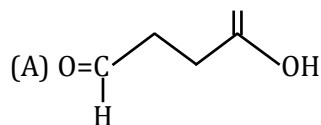
3. Which describes the best stereochemical aspects of the following reaction ?



- (A) Inversion of configuration occurs at the carbon undergoing substitution.
 (B) Retention of configuration occurs at the carbon undergoing substitution.
 (C) Racemization (loss of configuration) occurs at the carbon undergoing substitution.
 (D) The carbon undergoing substitution is not stereo genic.

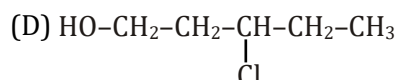
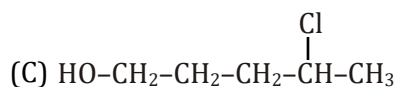
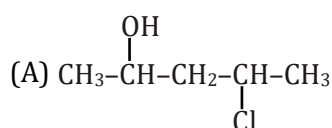
CAE192

4. $\text{HO}-\text{CH}_2-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{OH} \xrightarrow{\text{MnO}_2}$ (A), Product (A) is :

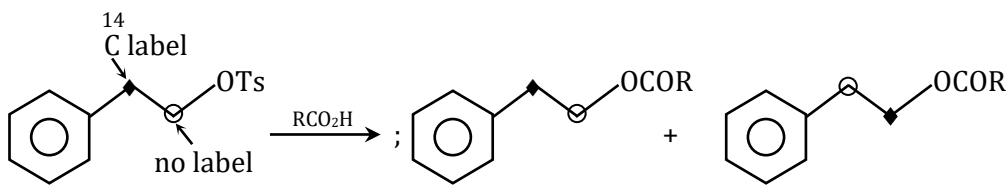
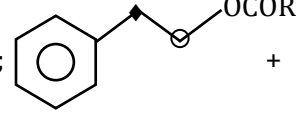
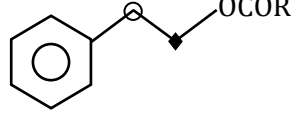


CAE193

5. $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}_2-\underset{\text{OH}}{\text{CH}}-\text{CH}_3 \xrightarrow[\text{Pyridine}]{\text{SOCl}_2} \text{(A)} \xrightarrow[\text{(H}_2\text{O)}]{\text{O}_3/\text{Zn}} \text{(B)} \xrightarrow[\text{C}_5\text{H}_9\text{ClO}]{\text{NaBH}_4} \text{(C)}$ Compound (C) is :

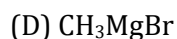
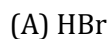
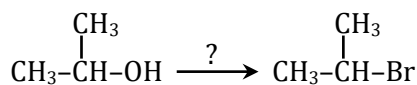


CAE194

6.  ;  + 
- (A) $\text{S}_{\text{N}}1$ (B) $\text{S}_{\text{N}}2$
(C) $\text{S}_{\text{N}}-\text{NGP}$ (D) $\text{S}_{\text{N}}-\text{Ar}$

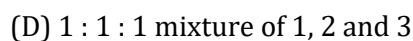
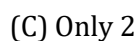
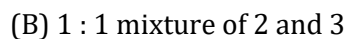
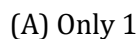
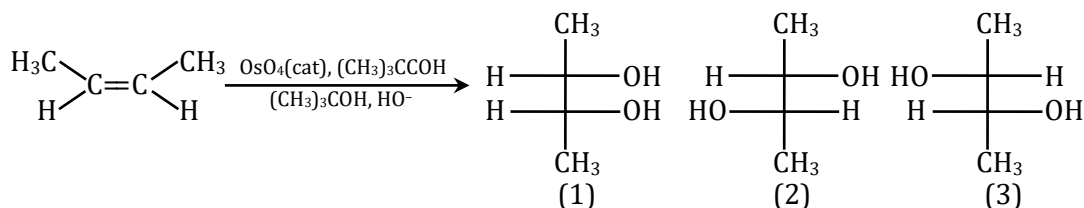
CAE181

7. Which is the best reagent to convert isopropyl alcohol to isopropyl bromide?



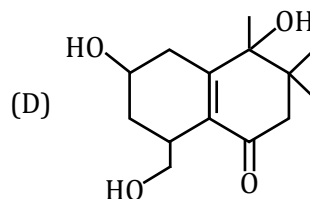
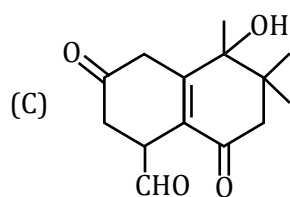
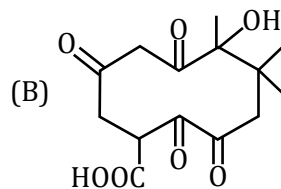
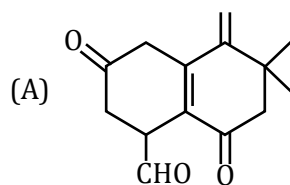
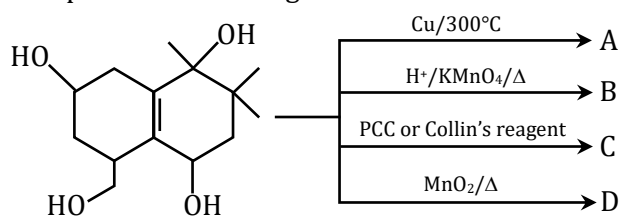
CAE182

8. What is the product of the following reaction?



CAE183

9. Complete the following reactions :



CAE185

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	D	D	C	C	C	C	B	A	ABCD